A Depth-Agile Vertical Array for Bidirectional Acoustic Data Recovery from Ocean-Bottom Seismometers

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***Abstract*—Ocean-bottom seismometers record for a year or more, but their data is unreachable until a ship recovers the instrument. Acoustically linked surface moorings solve this within a few kilometres. Their reach is limited by an effect transmit power cannot fix: a signal from a seafloor node reaches the relay twice, directly and by reflection from the sea surface, and the two copies interfere. At a reference site these arrivals are separated by 9.5° to 19.8° in elevation across 1–10 km, and the 60° transducer of a deployed moored observatory suppresses the reflection by under 3 dB. A 24-element vertical line array with a 4.23° beamwidth suppresses it by 13.6–32.9 dB, varying by less than 0.3 dB between seasons. Propagation is computed with the Bellhop Gaussian beam tracer on measured oceanography alone: sound-speed profiles from the NOAA World Ocean Atlas 2023, bathymetry from the GEBCO 2026 grid, a 34-year wave climate from Fisheries and Oceans Canada station C46206, and Hamilton's geoacoustic seabed values. Because such an array must be decoupled from surface wave motion it is carried on a buoyancy capsule that selects its park depth each dive from an onboard sound-speed cast. On measured oceanography this extends the usable radius from 3.22 km to 5.22 km in summer and 4.51 km to 5.21 km in winter—a 62% larger neighbourhood that does not move with season—and the same payload costs 62.4% less energy at its edge, a fraction independent of modem efficiency. An ablation confirms the array alone, at fixed depth, reaches only 3.07 km in winter—below the moored baseline—so the combination produces the result. The model reproduces the published operating ranges of two independently deployed systems without being fitted to either.**

***Index Terms*—Underwater acoustic communication, beamforming, multipath rejection, ocean-bottom seismometer, delay-tolerant networks, magnetoelectric antenna.**

I. INTRODUCTION

A broadband ocean-bottom seismometer (OBS) on the deep seafloor writes continuously for about a year. None of that data can be read until a vessel returns, fires an acoustic release and waits for the instrument to surface. Until then the science is inaccessible, failures go undetected, and the sampling strategy cannot respond to events.

Frye *et al.* addressed this with an acoustically linked moored observatory on the Nootka fault in 2004 [1]. A surface buoy carrying an acoustic modem and an Iridium terminal relayed data from two seafloor instruments for thirteen months at 5,400 bps, and it remains the reference architecture. Its limit is reach.

The paper reports approximately 4 km of acoustic telemetry, serving "sensors located over a 5–20 square kilometer area" [1]. An instrument outside that circle is not served slowly—it is not served at all, so the radius determines which instruments exist in the network rather than how fast any one link runs.

We present an array architecture that addresses this, with a buoyancy capsule as one embodiment. The contributions are array-side and therefore platform-independent, applying equally to a moored relay, a glider, or a vehicle harvesting a sensor field. First, we identify and measure the mechanism that sets the radius. The direct path and its surface reflection arrive 9.5°–19.8° apart, which a 60° transducer cannot separate and a 4.23° array suppresses by 13.6–32.9 dB. Second, we report a coverage result on measured oceanography: the usable radius rises from 3.22 to 5.22 km in summer and 4.51 to 5.21 km in winter, varying 0.2% between seasons against 40% for the baseline. Third, we show analytically that the resulting energy saving is independent of modem efficiency.

We do not claim that a single relay covers an arbitrary instrument field, nor that a drifting platform substitutes for a propelled vehicle where instruments are widely scattered. The claim is that the array hears better—and by reciprocity speaks better—at a given geometry. Quantities are tagged [V] verified, [C] computed, [S] simulated, [D] design choice, [A] assumption.

II. RELATED WORK

A. The site-matched baseline

Frye *et al.* [1] is uniquely appropriate: same fault, same water depth, published operating range and published delivered data. We model its receiver from its own stated hardware—"directional transducers with 60 degree beams," a source level of 190 dB, approximately 8 W transmit power, and transducers at the bottom of a 10 m chain below the universal joint [1]. A 60° cone subtends 0.842 sr, giving DI = $10 \log_{10}(4\pi/0.842)$ = 11.74 dB [C]. The baseline is therefore not omnidirectional, and modelling it as such would overstate the array's advantage.

B. Benchmark and link model

Send, Regier and Jones [2] fly gliders carrying acoustic modems in the California Current. Their site differs from ours, so we use their field measurements as a hardware benchmark rather than a head-to-head comparison: 75% delivery (1.33 transmissions per delivered packet) and 0.49 J per delivered byte [2]. Two findings bear directly here. Their Fig. 1 shows a near-surface shadow zone beyond 6 km for a transducer at 1,000 m and none at 3,200 m—published evidence that source depth determines shadowing. And their conclusion states that "in retrospect it might have been wiser to use a more directional transducer… as the increased directivity would increase the sound level received by the glider" [2]. This paper quantifies that recommendation.

We adopt the link budget and Rayleigh error model of Khan and Cho [3] unmodified, including their ambient-noise law, BPSK error rate under Rayleigh fading, and a 1,024-bit packet. They state that their nodes carry omnidirectional transducers and take DI as zero [3]. Introducing a non-zero DI at the relay, inside their own equations, is the intervention evaluated here.

C. Reconfigurable surfaces and mobile collection

Zhao *et al.* [4] establish that acoustic shadow zones are a binary failure mode—coverage below 20% regardless of radiated energy—and demonstrate a reconfigurable surface that restores it. Later work carries acoustic surfaces to tank-scale practice [19] and adds in-phase and quadrature modulation [20]. We adopt their gain law $DI = 10 \log_{10}(N)$, validated at $N = 2$ with a measured 2.69 dB against 3.01 dB predicted [4]. We cite them for physics but do not simulate their system: their surface is a mid-path reflector satisfying two legs at once, whereas the capsule is a two-point store-and-forward terminal with no middle position to optimise.

A parallel line collects from instruments using mobile vehicles, following the data-mule architecture of Shah *et al.* [21]. Hollinger *et al.* plan vehicle tours under a probabilistic communication model $P(x_v, x_n) = f(D(x_v, x_n))$ [15]; Khan *et al.* extend this with clustered path planning [16], and multihop variants relay between instruments [22]. These address *where the receiver should go*; the present work addresses *how well it hears when it arrives*. The two compose rather than compete—Hollinger *et al.* observe that as communication degrades their problem approaches the classical travelling-salesperson problem, because the vehicle must then approach each sensor closely [15]. The function f therefore governs how near a vehicle has to get. Notably the vehicles in both works carry omnidirectional modems with no beamforming, multipath rejection or depth selection, and the receiver evaluated here applies to them unchanged.

Underwater networking more broadly has been surveyed in [34]–[36]. Routing, localisation and energy management under acoustic constraints are each long-standing problems [37]–[41]. Optical and magnetic-induction links offer higher bandwidth over shorter ranges, and are studied as complements to acoustics rather than replacements for it [42]–[46]. Machine-learning methods have been applied across the stack [47], [48]. On the mobile side, vehicles have been used for hybrid optical–acoustic harvesting, energy-aware routing and multi-vehicle coordination [49]–[52], supported by acoustic positioning [53], and reconfigurable acoustic surfaces have been proposed for long-range and deep-sea use [54], [55]. Crossing the air–water interface directly has been demonstrated by acoustic-to-radio conversion, thermoacoustic coupling and backscatter [56]–[59], though all three remain short-range.

III. SYSTEM DESCRIPTION

The capsule is designed as a delay-tolerant, store-and-forward relay [23]. It is **not** intended for tsunami or earthquake early warning. At the surface, it receives its tasking before diving to a park depth determined from an onboard sound-speed cast. Once at depth, it beamforms on a seafloor node and buffers the received data. It then ascends and uplinks the stored data by satellite. The acoustic and radio links are separate systems that share one memory. The acoustic signal at 25 kHz is demodulated into bits and stored in memory. Hours later, those same bits are modulated onto a 1.6 GHz carrier for satellite transmission. There is no direct transduction between the acoustic and radio signals, and the design does not claim or rely on one.

We chose a vertical array because the two multipath arrivals we need to distinguish are separated in **elevation**. The receiver uses a 1 × 24 vertical line array operating at 25 kHz. With half-wavelength spacing, the elements are 3.00 cm apart, giving a 0.69 m aperture, a 4.23° beamwidth and 13.80 dB of broadside directivity. When the array is steered, the directivity falls to 11.25 dB [C], [S]. A 4 × 6 planar arrangement using the same 24 elements has only six elements vertically, resulting in 7.78 dB of directivity and a 16.92° beam [S]. This beam is too wide to resolve separations of roughly 19°, and the margin only worsens at longer ranges where the two arrivals converge further. The choice therefore comes down to the array's aspect ratio, rather than its element count or cost. The capsule itself is a slender vertical body measuring 0.20 m by 1.19 m, with a displacement of 37.4 L and a mass of 38.4 kg [C]. Its length is determined principally by the array aperture, and the capsule can be deployed by two people from a small vessel.

The capsule's vertical motion is controlled by a ball-screw-driven rigid piston [D]. The piston provides a 250 mL stroke, and the mechanism must deliver a peak thrust of 5.70 kN at a depth of 800 m. At a drivetrain efficiency of 0.505, the energy required is 2,455 J, or 0.68 Wh per cycle [C]. Deep-profiling floats of this class already operate routinely to depths of 6,000 m [31]. At 800 m, with a vertical speed of 0.15 m/s, a complete cycle takes 4 h 13 min [C]. Of this, 1 h 29 min is spent descending, 45 min on the acoustic session, another 1 h 29 min ascending, and 30 min in the surface window. The design cadence is one cycle per day [D], so the capsule is at the surface only 2% of the time, compared with 100% for a moored buoy.

IV. METHOD

A. Environment

All environmental inputs are measured data. Sound speeds were obtained from the NOAA World Ocean Atlas 2023 one-degree statistical means at 49.5° N, 128.5° W [7], converted by the Mackenzie nine-term equation [8], and standard ocean-acoustic relations from Urick [24]. In summer $c(0) = 1501.33$ m/s against $c(2372) = 1495.81$ m/s with an axis at 400 m, and in winter $c(0) = 1479.81$ m/s against $c(2372) = 1495.61$ m/s with an axis at 500 m. **The surface-to-seabed relationship inverts between seasons**, and this drives the behaviour reported in Section V-B.

Bathymetry from GEBCO 2026 [9] gives 2,341 m with a mean slope of 0.2 m/km over ±36 km, independently confirming Frye's reported depth to 31 m from an unconnected source and justifying the range-independent model. Sea state from DFO station C46206 [10], 203,885 quality-controlled records 1988–2022, gives significant wave heights of 1.95 m (median), 4.39 m (P95) and 5.73 m (P99); wave orbital amplitude decays as $\exp(-2\pi z/L)$, and requiring displacement below $\lambda/10$ at 25 kHz places the beamforming floor at **329 m** at the 95th-percentile sea state [C]. Seabed properties are Hamilton's silt values: 1,767 kg/m^3, 1,623 m/s, 0.673 dB/kHz [11].

B. Propagation

The underwater acoustic channel varies with range, frequency and depth, with multipath propagation and limited bandwidth being its defining constraints [25]. We model propagation using the Bellhop Gaussian beam tracer [12], which returns the amplitude, delay, source angle, **receiver-side arrival angle** and bounce counts for each eigenray. The receiver-side arrival angle is particularly important because each ray can reach the receiver from a different direction. The array's beam pattern must therefore be evaluated at the actual arrival angle of each ray, rather than at one assumed direction. This allows directivity and multipath rejection to be accounted for through the geometry of the propagation paths and the array, rather than being added to the link budget as a constant. It also explains why a 60° transducer and a 4.23° array produce different results even when the ray

paths are identical. We calculate transmission loss by coherently summing the gain-weighted arrivals and averaging the result over 25 samples across the 4 kHz band. Thorpe absorption is applied to each arrival over its actual path length.

Every result rests on two tests kept distinct. The first is geometric: does a direct ray arrive at all? A shadowed link is not recoverable by raising source level [4]. The second is the link budget: is that ray strong enough to carry data? This is recoverable by power or coding. Conflating them obscures which constraint binds.

C. Validation

The ray engine was validated against the exact closed form for straight rays in an isovelocity ocean. For a source at 200 m and receiver at 2,371 m at 3 km, the analytic direct path is 3,703.1 m at +35.892°; Bellhop returns 3,703.1 m at +35.892°—an error of 0.0000° and 0.00 m [S]. A second validation appears in the results: **the model reproduces the published operating range of two independently deployed systems without being fitted to either.** Frye's link fails beyond 4 km, and his paper reports approximately 4 km [1]. The Send glider degrades beyond 3 km, and its paper reports 3.5–5.5 km typical [2].

V. RESULTS

A. The array rejects the surface echo

A seafloor node transmits once, but its signal reaches the relay twice—directly and after reflecting from the sea surface. The copies arrive milliseconds apart at comparable amplitude and interfere, corrupting packets. Measured delay profiles in shallow water confirm surface-reflected arrivals at amplitudes comparable to the direct path [26]. Their angular separation is given in Table I.

TABLE I
Surface-reflection suppression [S]. Both arrivals lie within the main lobe of a 60° beam, whose −3 dB points are at ±30°.

Range	Separation	Moored 60°	Array 4.23°
1.0 km	15.04°	< 3 dB	13.6 dB
2.0 km	19.76°	< 3 dB	22.5 dB
3.0 km	19.18°	< 3 dB	24.0 dB
4.0 km	17.28°	< 3 dB	26.5 dB
5.5 km	14.47°	< 3 dB	23.6 dB
7.0 km	12.32°	< 3 dB	17.7 dB
10.0 km	9.53°	< 3 dB	32.9 dB

Winter agrees with summer to within 0.3 dB at every range [S]: the separation is set by the receiver's height above the seabed and its distance from its own surface image, neither of which is seasonal. The mechanism is likewise frequency-independent, since at half-wavelength spacing 24 elements give 4.23° at any frequency with only the aperture changing. It does have a ceiling: at 20 km the paths converge to 0.21°, below the beamwidth, so the mechanism is bounded near 12–15 km regardless of transmit power.

B. Larger coverage, unchanged by season

TABLE II
Usable radius by configuration and season [S].

Configuration	Summer	Winter	Swing
Moored, 10 m, 60°	3.22 km	4.51 km	40%
Array alone, fixed 800 m	3.94 km	3.07 km	28%
Depth-agile array	5.22 km	5.21 km	0.2%

The depth-agile radius does not move with season. The moored baseline varies by 40% because its transducer is fixed at 10 m while the surface layer is not; a relay that re-solves its depth each cycle lands within 0.01 km of the same radius in both. A moored field must therefore be sized for its worst season and is over-provisioned for the rest of the year, where a capsule field is sized once. Coverage area scales with radius squared: 32.5 → 85.6 km^2 in summer (2.63×), 63.9 → 85.3 km^2 in winter (1.33×). In the formulation of [15] this is the probabilistic neighbourhood around each instrument, which those methods take as a fixed input.

The 3.22 km baseline figure is itself a validation: Frye reports approximately 4 km and 5–20 km^2 of coverage [1], and we obtain 3.22 km and 32.5 km^2 from physics with nothing tuned to match.

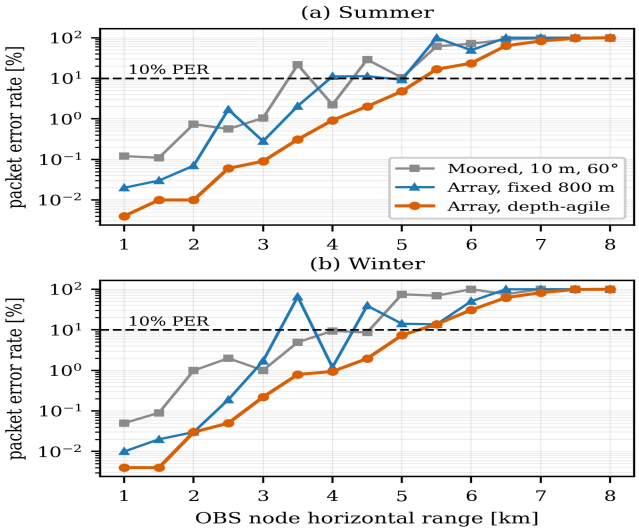


Fig. 1. Packet error rate against node range, both seasons. The depth-agile array stays below the 10% usability threshold to 5.22 km in summer and 5.21 km in winter; the moored baseline reaches 3.22 km and 4.51 km respectively. The fixed-depth array falls below the baseline in winter.

C. Less energy for the same data

Delivering a payload requires retransmitting failed packets. Using the baseline's own parameters—500 kB/day for 395 days, 193 MB total [1]—gives Table III.

TABLE III
Energy to deliver 193 MB over a 13-month deployment [S].

Case	Moored	Array	Saving
5 km, summer	1,804.6 kJ	1,698.8 kJ	5.9%
6 km, summer	5,625.7 kJ	2,113.6 kJ	62.4%
4 km, winter	1,785.4 kJ	1,633.2 kJ	8.5%
5 km, winter	6,384.6 kJ	1,746.1 kJ	72.7%
6 km, winter	no link	2,341.6 kJ	—

The mechanism is the retransmission count: 3.48 against 1.31 transmissions per packet at 6 km summer and 3.95 against 1.08 at 5 km in winter [S]. **This is independent of modem efficiency.**

Energy is bits × (transmissions per packet) / η, so in the ratio

$$saving = 1 - (tx_{array} / tx_{moored})$$

η appears in both terms and cancels exactly. Verified numerically at 500, 1,000, 1,500 and 2,000 bits per joule, the saving is 62.4% in every case to four significant figures. Only the absolute joule figures scale. Within 4 km the saving is under 2% and operationally negligible—the gain lives at the edge of the neighbourhood, where the baseline's retransmission count climbs steeply.

D. The satellite uplink and the reverse link

The capsule surfaces to uplink, and the antenna it carries is constrained by the hull rather than by the link budget. At 1.616 GHz over a 1,400 km slant with 7 W transmit and −10 dB/K satellite G/T, free-space path loss is 159.54 dB. A +2 dBi monopole then yields E_b/N_0 = 35.71 dB against a QPSK requirement of 10.5 dB [C]. **The margin is 25.21 dB, so the antenna may be as poor as −23.2 dBi.** We are not aware of a prior statement of this bound for a surfacing acoustic relay. It matters because it admits a conformal element with no coaxial penetration of the pressure boundary, in place of a protruding whip. Magnetoelectric antennas are one candidate class: they resonate mechanically rather than electrically, so their frequency does not shift when the element is wetted [13], and reported devices are conformal at roughly 1% of the electromagnetic wavelength [14], [27], [28].

The acoustic link is bidirectional in operation, since the capsule must deliver commands and retransmission requests to the instrument, as the reference glider system also reports [2]. An aperture is reciprocal, so the same 24 elements form a 4.23° transmit beam with the same 13.80 dB of directivity. Two consequences follow. Concentrating that gain toward the instrument is a power ratio of 24×, so the same received level requires 176.2 dB rather than 190 dB. And a transmit beam that does not illuminate the sea surface never creates the reflection the forward link has to reject. The instrument listens with one omnidirectional hydrophone and cannot discriminate spatially, so the array is the only element in the link able to improve the downlink. Both consequences follow from reciprocity and Table I rather than from separate simulation.

VI. DISCUSSION

A. The park-depth window

Park depth is bounded on both sides by measured physics, and the upper bound is not fixed. The floor is 329 m, set by the 95th-percentile sea state [10]. Below it the array cannot hold phase against wave orbital motion. Profile inversion from in-situ measurement is an active field [30], and the capsule derives its depth from a cast rather than from climatology. The ceiling is geometric reach, which falls as the relay descends because water below the axis is faster and a deeper source needs a steeper launch angle (Table IV).

TABLE IV
Geometric reach to a seabed node against park depth, summer [C].

Park depth	400 m	800 m	1,200 m	1,600 m	2,000 m
Max reach	20.1 km	17.9 km	15.1 km	12.0 km	8.3 km

Shadow-zone extent is conventionally established by ray simulation [32]. Snell ray integration and Bellhop agree on this boundary [S]. **The ceiling moves with target range**—about 2,000 m for an 8 km node, about 450 m for a 20 km node—so no single fixed depth is simultaneously optimal, which is the physical basis for depth agility. Park depth is not the sound-channel axis at 400–500 m; for a seabed target what matters is distance from the receiver's own surface image.

B. Neither element is sufficient alone

A natural objection is to mount an array on the existing mooring. Table II answers it: the same 24 elements at a fixed 800 m reach 3.94 km in summer and **3.07 km in winter, below the moored baseline's 4.51 km**. An array at an unsuitable depth is worse than no array at all, so angular resolution and the ability to position it are each insufficient in isolation. This is also the closest available analogue to a stationary-versus-mobile comparison within a two-point geometry.

C. Scope

Radius is not a data-rate result: total data recoverable from any single instrument is unchanged, and a relay at 6 km moves the same bits per second as one at 3 km. What the radius determines is the geometry a deployment can adopt. Instrument spacing in acoustically telemetered deployments is constrained by relay reach rather than chosen independently of it—the reference deployment placed its instruments within 3 km against a stated 4 km limit [1]. Published spacings span a wide range, from 5–10 km for academic refraction profiles to 110–160 km for mantle-scale passive arrays such as UPFLOW [18]. A usable radius of 5.22 km brings the refraction-profile regime within reach of a single relay position; at mantle scale no acoustic relay is applicable.

Within 4 km, where the reference deployment's own instruments sit, both configurations achieve packet error rates near 1% and the energy saving is under 2%. We do not claim a drifting relay substitutes for a propelled vehicle: where instruments are widely scattered, an AUV data mule is the appropriate architecture [15], [16], and the array-side results here transfer to it unchanged.

VII. LIMITATIONS

Three assumptions [A] remain in the acoustic model. WOA23 terminates at 2,200 m at this cell, so the final 172 m is linearly extrapolated—under 16% of the column, in near-isovelocity water. The 60° baseline transducer is modelled with a flat top and −20 dB skirt, which is why Table I reports "< 3 dB" rather than a specific value. Drivetrain efficiencies other than the THK ball-screw figure [5] are estimates. The RF budget uses standard Iridium-class values [A], and the outage duration in Section V-D is read from the published daily-transmission record [1, Fig. 6].

Four limitations of method apply. Bellhop is a two-dimensional ray model over a horizontally uniform ocean, without currents, internal waves, roughness or out-of-plane propagation; the measured 0.2 m/km slope [9] supports the range-independent approximation here. The Rayleigh fading model requires 33.9 dB SNR for 10% packet error rate, whereas beamforming partly creates a Rician channel, so our figures are conservative. The array is steered exactly at the direct arrival, where a real system estimates that angle with error; on a 4.23° beam a degree or two costs several decibels of main-lobe gain, and adaptive weighting (MVDR or LCMV) is the standard mitigation. Finally, beyond approximately 15 km the two arrivals converge below the beamwidth, so the architecture targets dense local clusters.

VIII. CONCLUSION

The reach of an acoustically linked seafloor observatory is limited by an interference mechanism that transmit power cannot address. At the Nootka fault the direct path and its surface reflection arrive 9.5° to 19.8° apart across 1–10 km, and the 60° transducer of a deployed moored observatory suppresses the reflection by under 3 dB, where a 24-element vertical array with a 4.23° beamwidth suppresses it by 13.6 to 32.9 dB, varying by less than 0.3 dB between seasons.

Because such an array must be decoupled from surface wave motion—a floor of 329 m at the 95th-percentile sea state—it is carried on a buoyancy capsule that selects its park depth each dive. This extends the usable radius from 3.22 km to 5.22 km in summer and 4.51 km to 5.21 km in winter, the depth-agile radius varying by 0.2% against 40% for the baseline, and the same payload costs 62.4% less energy at the edge of that neighbourhood. The satellite uplink carries 25.21 dB of margin, admitting a conformal antenna of −23.2 dBi and so a hull with no coaxial penetration.

An ablation establishes that neither element suffices alone, the same array at a fixed 800 m reaching only 3.07 km in winter. The contribution is array-side and platform-independent: multipath rejection, retransmission cost and reverse-link focusing hold on any platform capable of carrying the array, and in the formulation used by AUV path-planning work [15], [16] the receiver enlarges the communication neighbourhood those methods take as a fixed input. Future work will replace the climatological profile with a measured cast, implement adaptive beamforming against pointing error, and evaluate the architecture on a surveyed instrument geometry.

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